

TALEEF TAMSAL

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SUMMARY

Software engineer building production AI, backend, and healthcare systems. Sole developer of a production healthcare platform and LLM safety researcher with four first-author papers, including a NeurIPS 2026 workshop oral.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash, R

AI & ML: PyTorch, Hugging Face, PEFT, LangGraph, RAG, AI agents, MCP, LLM evaluation, scikit-learn

Backend & Web: Node.js, Express, FastAPI, Spring Boot, Next.js, React, REST APIs, PostgreSQL, Redis

Cloud & DevOps: Docker, GitHub Actions, CI/CD, Azure, AWS, Linux, SLURM, Playwright, OpenTelemetry

Healthcare & Certs: FHIR R4, CQL, CMS eCQMs, QRDA, AWS AI Practitioner, AWS Cloud Practitioner

EXPERIENCE

Medical Informatics Engineering

May 2026 – Present

Software Developer

- Sole developer of WorkWell Measure Studio, a clinical quality measure platform for WebChart, retiring its Java/Spring Boot backend for **TypeScript** across **500+ pull requests** and **3,268 automated tests**.
- Executes CMS's own FHIR measure artifacts behind a permanent CI gate of **455 of 455** official MADiE test cases across **9 measures**, agreeing with HAPI's independent engine on **362 of 387** cases.
- Scaled a **20,000-patient** sandbox to **120,000 nightly evaluations**, making official execution **10 to 16x faster** through batching and cutting a timed-out CSV export from 60s to **7.7 s**.
- Integrated live **HL7 FHIR R4** data over SMART Backend Services and shipped a **13-tool read-only MCP server**, CDS Hooks cards, and MeasureReport and QRDA exports validating at zero errors.

Purdue University

Sep 2025 – May 2026

Graduate Researcher, AI Safety & LLM Evaluation

- Fine-tuned and evaluated **25 LLM configurations** across five model families (4B to 14B) with full fine-tuning, LoRA, QLoRA, and 4-bit quantization via **SLURM** on A100 and A30 GPUs.
- Showed via equivalence testing that **LoRA and full fine-tuning degrade jailbreak safety equivalently**, across nearly **600 attack-success measurements** (PAIR, AutoDAN, HarmBench).
- Ran a **750-sample blind human annotation** study exposing a **45.7% false-positive rate** in an LLM judge vs. **8.6%** for a calibrated classifier, inflating measured degradation **about 3x**.

Riccle (early-stage e-commerce startup)

Sep 2025 – May 2026

Product Associate, Operations & Analytics

- Consolidated order, inventory, payment, and customer data from **four trackers into one** source and built recurring KPI reporting with SQL and Python, cutting **about 6 hours a week**.

Medical Informatics Engineering

May 2025 – Aug 2025

Software Development Intern

- Architected a containerized **RAG platform** with hybrid search and reranking over **3,000+** issues and MCP and SSE interfaces, serving **25 concurrent users and AI agents under 500 ms**.

PUBLICATIONS (first author, 2026)

- Your Judge Is a Confound*, **NeurIPS 2026 JUDGE Workshop (oral)**, AdvML-Frontiers @ COLM 2026
- Adaptation, Not Algorithm*, PEFT vs. full fine-tuning safety, **AACL-IJCNLP 2026 Findings**
- Two SemEval-2026 system papers in the **ACL Anthology**, one ranking **6th of 26** on generation

PROJECTS

JobOps Copilot, Multi-Tenant AI Agent Platform | [GitHub](#) | [Live](#)

May 2026 – Present

- Built a Next.js, TypeScript, and Python platform running stateful **LangGraph** agents with pgvector RAG and MCP tool use, plus tenant isolation, PII redaction, and **CI/CD quality gates**.

EDUCATION

Purdue University, M.S., Computer Science, GPA 4.0

Aug 2024 – May 2026

LUMS, B.S., Computer Science

Aug 2020 – May 2024